Chapter 7. Advanced Group Theory

Isomorphism theorems, series of groups, Sylow theorems and applications, free abelian groups, free groups, presentations

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7.1 Isomorphism theorems (§34)

Motivation. The fundamental homomorphism theorem has several useful consequences that describe how subgroups and factor groups interact. They are the bookkeeping tools for the composition series and Sylow arguments that follow.

Theorem 7.1 (34.2, first isomorphism theorem)

Let $\phi : G \to G'$ be a homomorphism with kernel K and $\gamma_K : G \to G/K$ the canonical map. There is a unique isomorphism $\mu : G/K \to \phi[G]$ with $\phi(x) = \mu(\gamma_K(x))$ for all x.

Proof. Existence is Theorem 14.11.

Uniqueness: $\mu(xK) = \mu(\gamma_K(x))$ must equal $\phi(x)$. ■

Lemma 7.2 (34.3, correspondence)

Let $N \trianglelefteq G$ and $\gamma : G \to G/N$. Then $L \mapsto \gamma[L]$ is a bijection from the normal subgroups of G containing N onto the normal subgroups of G/N.

Proof. $\gamma[L]$ is normal in G/N (Theorem 15.16). It is onto: for $\bar{L} \trianglelefteq G/N$, $L = \gamma^{-1}[\bar{L}]$ is normal, contains N, and $\gamma[L] = \bar{L}$ (γ is onto). One-to-one: if $N \leq L$, then $\gamma^{-1}[\gamma[L]] = LN = L$; so L is recovered from $\gamma[L]$. ■

The same proof shows the correspondence for all subgroups (not only normal ones).

Lemma 7.3 (34.4)

If $N \trianglelefteq G$ and $H \leq G$, then $H \vee N = HN = NH$ (the subgroup generated by $H \cup N$). If moreover $H \trianglelefteq G$, then $HN \trianglelefteq G$.

Proof. $hn = (hnh^{-1})h \in NH$ and similarly $NH \subset HN$. HN is a subgroup: $(h_1n_1)(h_2n_2) = h_1h_2(h_2^{-1}n_1h_2)n_2$ and $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}(hn^{-1}h^{-1})$. Any subgroup containing H and N contains HN, so $HN = H \vee N$. If H is also normal, $g(hn)g^{-1} = (ghg^{-1})(gng^{-1}) \in HN$. ■

Theorem 7.4 (34.5, second isomorphism theorem)

If $H \leq G$ and $N \trianglelefteq G$, then $H \cap N \trianglelefteq H$ and $HN/N \cong H/(H \cap N)$.

Proof. $\phi : H \to HN/N$, $\phi(h) = hN$, is a homomorphism; it is onto since $hnN = hN$; its kernel is $H \cap N$. Apply Theorem 7.1. ■

Theorem 7.5 (34.7, third isomorphism theorem)

If $K \leq H$ are normal subgroups of G, then $H/K \trianglelefteq G/K$ and $G/H \cong (G/K)/(H/K)$.

Proof. $\phi : G/K \to G/H$, $\phi(xK) = xH$, is well defined ($K \leq H$), an onto homomorphism, with kernel $\{xK : x \in H\} = H/K$. Apply Theorem 7.1. ■

7.2 Series of groups (§35)

Motivation. Just as integers factor into primes, a finite group can be broken by a chain of normal subgroups into simple factor groups. The Jordan–Hölder theorem says these "prime factors" are unique, and the notion of a solvable group (all factors abelian) is exactly what Galois theory will need for solving equations by radicals.

A **subnormal series** of G is a chain $\{e\} = H_0 < H_1 < \cdots < H_n = G$ with $H_i \trianglelefteq H_{i+1}$; a **normal series** has all $H_i \trianglelefteq G$. A **refinement** inserts further groups. Two series are **isomorphic** if there is a one-to-one correspondence between their factor groups H_{i+1}/H_i with corresponding factors isomorphic. A **composition series** is a subnormal series with all factors simple; a **principal series** is a normal series with no proper normal refinement. G is **solvable** if it has a composition series with abelian (hence cyclic of prime order) factors, equivalently a subnormal series with abelian factors.

Lemma 7.6 (35.10, Zassenhaus)

Let $H, K \leq G$, $H^* \trianglelefteq H$, $K^* \trianglelefteq K$. Then
(1) $H^*(H \cap K^*) \trianglelefteq H^*(H \cap K)$;
(2) $K^*(H^* \cap K) \trianglelefteq K^*(H \cap K)$;
(3) $$\frac{H^*(H \cap K)}{H^*(H \cap K^*)} \cong \frac{K^*(H \cap K)}{K^*(H^* \cap K)} \cong \frac{H \cap K}{(H^* \cap K)(H \cap K^*)}.$$

Proof. $H \cap K$ normalizes H^* (as $H^* \trianglelefteq H$), so $H^*(H \cap K)$ is a subgroup; similarly the other products. $H^* \cap K \trianglelefteq H \cap K$ (intersect the normal H^* of H with $H \cap K$), likewise $H \cap K^* \trianglelefteq H \cap K$, so $L = (H^* \cap K)(H \cap K^*) \trianglelefteq H \cap K$. Define $\phi : H^*(H \cap K) \to (H \cap K)/L$ by $\phi(hx) = xL$ ($h \in H^*$, $x \in H \cap K$). Well defined: if $hx = h'x'$, then $x'x^{-1} = h'^{-1}h \in H^* \cap (H \cap K) = H^* \cap K \subset L$. Homomorphism: $(h_1x_1)(h_2x_2) = h_1(x_1h_2x_1^{-1})x_1x_2$ with $x_1h_2x_1^{-1} \in H^*$. It is onto, and $\ker\phi = \{hx : x \in L\} = H^*L = H^*(H \cap K^*)$ (since $H^* \cap K \subset H^*$). This proves (1) and the isomorphism of the first and third groups in (3); by symmetry ($H \leftrightarrow K$) we get (2) and the second isomorphism. ∎

Theorem 7.7 (35.11, Schreier)

Two subnormal (normal) series of G have isomorphic refinements.

Proof. Let $\{e\} = H_0 < \cdots < H_n = G$ and $\{e\} = K_0 < \cdots < K_m = G$. Insert $H_{i,j} = H_i(H_{i+1} \cap K_j)$ ($0 \leq j \leq m$) between $H_i = H_{i,0}$ and $H_{i+1} = H_{i,m}$, and $K_{j,i} = K_j(H_i \cap K_{j+1})$ between K_j and K_{j+1}. By Lemma 7.6 (with $H = H_{i+1}$, $H^* = H_i$, $K = K_{j+1}$, $K^* = K_j$), $H_{i,j} \trianglelefteq H_{i,j+1}$, $K_{j,i} \trianglelefteq K_{j,i+1}$, and $H_{i,j+1}/H_{i,j} \cong K_{j,i+1}/K_{j,i}$. Both refined series have nm steps, matched by $(i,j) \leftrightarrow (j,i)$. (Repeated groups give trivial factors, which correspond to each other.) For normal series all groups are normal in G. ∎

Theorem 7.8 (35.15, Jordan–Hölder)

Any two composition (principal) series of G are isomorphic.

Proof. By Theorem 7.7 they have isomorphic refinements. A refinement of a composition series only adds repeated groups (factors are simple, so no proper normal subgroup can be inserted, Theorem 15.18), i.e. trivial factors. Deleting the trivial factors on both sides leaves an isomorphism of the original series. ∎

Theorem 7.9 (35.16)

If G has a composition (principal) series and N is a proper normal subgroup, there is a composition (principal) series containing N.

Proof. The series $\{e\} < N < G$ and the given series have isomorphic refinements. The refinement of the composition series has only simple or trivial factors; so does the isomorphic refinement of $\{e\} < N < G$. Removing repetitions from the latter gives a composition series containing N. ■

Example 7.1

$\{e\} < A_n < S_n$ is a composition series for $n \geq 5$, so S_n is not solvable. The center $Z(G)$ and the ascending central series $Z_0 = \{e\} \leq Z_1 = Z(G) \leq Z_2 \leq \cdots$ $(Z_{i+1}/Z_i = Z(G/Z_i))$ are defined; $\mathbb{Z}_{60}$ has composition factors $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_5$ in every composition series.

7.3 Sylow theorems (§36)

Motivation. Lagrange's theorem says the order of a subgroup divides $|G|$, but the converse fails (A_4 has no subgroup of order 6). The Sylow theorems give the strongest partial converse: subgroups of every maximal prime-power order exist, are all conjugate, and their number is tightly constrained. The proofs are applications of group actions and orbit counting.

A **p-group** is a group in which every element has order a power of p.

Theorem 7.10 (36.1)

If $|G| = p^n$ and X is a finite G-set, then $|X| \equiv |X_G| \pmod{p}$, where X_G is the set of fixed points.

Proof. X is the disjoint union of orbits; orbits of one element are the fixed points, and every other orbit has size $(G : G_x)$, a divisor of p^n larger than 1 (Theorem 16.16), hence divisible by p. ■

Theorem 7.11 (36.3, Cauchy)

If p is prime and $p \mid |G|$, then G has an element of order p, hence a subgroup of order p.

Proof. Let $X = \{(g_1, \ldots, g_p) \in G^p : g_1 g_2 \cdots g_p = e\}$; $|X| = |G|^{p-1}$ (g_p is determined by the others), divisible by p. $\mathbb{Z}_p$ acts on X by cyclic shifts (if $g_1 \cdots g_p = e$ then $g_2 \cdots g_p g_1 = g_1^{-1} e g_1 = e$). The fixed points are the $(g, \ldots, g)$ with $g^p = e$. By Theorem 7.10 their number is $\equiv 0 \pmod{p}$; it is at least 1 ($g = e$), so there is $g \neq e$ with $g^p = e$. ■

Corollary 7.12 (36.4)

A finite group is a p-group iff its order is a power of p.

Proof. If a prime $q \neq p$ divides $|G|$, Cauchy gives an element of order q. The converse is Lagrange. ■

A **Sylow p-subgroup** is a maximal p-subgroup. The **normalizer** of H is $N[H] = \{g : gHg^{-1} = H\}$, the largest subgroup in which H is normal.

Lemma 7.13 (36.6)

If H is a p-subgroup of a finite group G, then $(N[H] : H) \equiv (G : H) \pmod{p}$.

Proof. H acts on the set $\mathcal{L}$ of left cosets of H by left multiplication. xH is fixed iff $hxH = xH$ for all $h \in H$, iff $x^{-1}hx \in H$ for all h, iff $x \in N[H]$ ($x^{-1}Hx = H$, by finiteness). So the fixed points are the $(N[H] : H)$ cosets of H in $N[H]$; apply Theorem 7.10. ■

Corollary 7.14 (36.7)

If H is a p-subgroup and $p \mid (G : H)$, then $N[H] \neq H$.

Proof. $(N[H] : H) \equiv 0 \pmod{p}$ and is at least 1, so it is at least p. ■

Theorem 7.15 (36.8, first Sylow theorem)

Let $|G| = p^n m$, $n \geq 1$, $p \nmid m$. Then (1) G has a subgroup of order p^i for each $1 \leq i \leq n$;
(2) every subgroup H of order p^i, $i < n$, is a normal subgroup of a subgroup of order p^{i+1}.

Proof. Cauchy gives order p. Let $|H| = p^i < p^n$. Then $p \mid (G : H)$, so $p \mid (N[H] : H) = |N[H]/H|$ (Lemma 7.13). By Cauchy, $N[H]/H$ has a subgroup K of order p; its inverse image under $N[H] \to N[H]/H$ has order p^{i+1} and contains H as a normal subgroup. Induction gives (1). ■

So the Sylow p-subgroups of G are exactly the subgroups of order p^n.

Theorem 7.16 (36.10, second Sylow theorem)

Any two Sylow p-subgroups P_1, P_2 of a finite group are conjugate.

Proof. P_1 acts on the left cosets of P_2; their number $(G : P_2)$ is prime to p, so by Theorem 7.10 some xP_2 is fixed: $P_1 x P_2 = xP_2$, so $x^{-1}P_1 x \leq P_2$, with equality by orders. ■

Theorem 7.17 (36.11, third Sylow theorem)

If $p \mid |G|$, the number of Sylow p-subgroups is $\equiv 1 \pmod{p}$ and divides $|G|$.

Proof. Let P be a Sylow subgroup; P acts by conjugation on the set $\mathcal{S}$ of Sylow p-subgroups. If Q is fixed, $P \leq N[Q]$, so P and Q are Sylow subgroups of $N[Q]$, conjugate there (Theorem 7.16); as $Q \trianglelefteq N[Q]$, $P = Q$. So there is exactly one fixed point and $|\mathcal{S}| \equiv 1$. G acts transitively on $\mathcal{S}$ by conjugation (Theorem 7.16), so $|\mathcal{S}| = (G : N[P])$ divides $|G|$. ■

7.4 Applications of the Sylow theory (§37)

Motivation. The Sylow constraints are often enough to determine a group completely or to show that no group of a given order is simple. This section shows the typical arguments, for example that every group of order pq with $p < q$, $q \not\equiv 1 \pmod{p}$, is cyclic.

Theorem 7.18 (37.1)

Every finite p-group is solvable.

Proof. By Theorem 7.15(2) applied repeatedly there is a chain $\{e\} = H_0 < H_1 < \cdots < H_n = G$, $|H_i| = p^i$, $H_i \trianglelefteq H_{i+1}$; the factors have order p, hence are abelian. ■

Theorem 7.19 (37.4)

A finite nontrivial p-group has nontrivial center.

Proof. Let G act on itself by conjugation; the fixed points form the center $Z(G)$. By Theorem 7.10, $|Z(G)| \equiv |G| \equiv 0 \pmod{p}$, and $e \in Z(G)$. ■

Lemma 7.20 (37.5)

If $H, K \trianglelefteq G$, $H \cap K = \{e\}$ and $HK = G$, then $G \cong H \times K$.

Proof. For $h \in H$, $k \in K$, $hkh^{-1}k^{-1} \in H \cap K$ (it is $(hkh^{-1})k^{-1} \in K$ and $h(kh^{-1}k^{-1}) \in H$), so $hk = kh$. Then $(h, k) \mapsto hk$ is a homomorphism $H \times K \to G$, onto, with kernel $\{(h, h^{-1}) : h \in H \cap K\} = \{(e, e)\}$. ■

Theorem 7.21 (37.6)

Every group of order p^2 is abelian.

Proof. $Z(G) \neq \{e\}$. If $Z(G) \neq G$, then $|Z(G)| = p$ and $G/Z(G)$ is cyclic, generated by $aZ(G)$; then every element is $a^i z$ with $z \in Z(G)$, and such elements commute, so G is abelian, $Z(G) = G$, a contradiction. ■

Theorem 7.22 (37.7)

If $p < q$ are primes, a group of order pq has a unique, hence normal, subgroup of order q; it is not simple. If $q \not\equiv 1 \pmod{p}$, then G is cyclic.

Proof. The number of Sylow q-subgroups divides pq, is $\equiv 1 \pmod{q}$, so is 1 (as $p < q$). If $q \not\equiv 1 \pmod{p}$, the number of Sylow p-subgroups, which divides q and is $\equiv 1 \pmod{p}$, is also 1. Both are normal, intersect trivially, and their product has order pq; by Lemma 7.20, $G \cong \mathbb{Z}_p \times \mathbb{Z}_q \cong \mathbb{Z}_{pq}$. ■

Lemma 7.23 (37.8)

For finite subgroups H, K of G, $|HK| = |H||K|/|H \cap K|$.

Proof. HK is the union of the cosets hK, $h \in H$, each of size $|K|$; $hK = h'K$ iff $h^{-1}h' \in H \cap K$, so there are $(H : H \cap K)$ distinct ones. ■

Example 7.2

No group of order 15, 20, 30, 36, 48, or 255 is simple. For order 30: if neither $n_3 = 1$ nor $n_5 = 1$, then $n_3 = 10$, $n_5 = 6$, giving 20 elements of order 3 and 24 of order 5, too many. For order 36: if $n_3 = 4$, the action on the 4 Sylow 3-subgroups gives a homomorphism to S_4 with a nontrivial kernel (as $36 > 24$). For order 48: two Sylow 2-subgroups H, K of order 16 have $|H \cap K| \geq 8$ by Lemma 7.23, and $N[H \cap K]$ contains H and K, so has order 48.

7.5 Free abelian groups (§38)

Motivation. Free abelian groups play the role of vector spaces with a basis among abelian groups. Every finitely generated abelian group is a quotient of $\mathbb{Z}^n$, and choosing good bases for $\mathbb{Z}^n$ and its subgroups gives a full proof of the fundamental theorem stated in §11.

Theorem 7.24 (38.1)

For a subset X of a nonzero abelian group G the following are equivalent:
(1) each nonzero $a \in G$ is uniquely $a = n_1x_1 + \cdots + n_rx_r$ with distinct $x_i \in X$ and $n_i \neq 0$;
(2) X generates G, and $n_1x_1 + \cdots + n_rx_r = 0$ for distinct $x_i \in X$ only if all $n_i = 0$.

Proof. (1)⇒(2): if $\sum n_ix_i = 0$ with some $n_i \neq 0$, then for such an x_j, $x_j = x_j + \sum n_ix_i$ gives two different expressions of x_j (after collecting terms). (2)⇒(1): two expressions of a subtract to a relation, whose coefficients must vanish. ■

Such an X is a **basis**, and G is **free abelian**.

Theorem 7.25 (38.5)

A free abelian group with a basis of r elements is isomorphic to $\mathbb{Z}^r$.

Proof. $\sum n_ix_i \mapsto (n_1, \ldots, n_r)$ is an isomorphism. ■

Theorem 7.26 (38.6)

If $G \neq \{0\}$ is free abelian with a finite basis, every basis is finite and all bases have the same number of elements (the **rank**).

Proof. If X is a basis with r elements, $G/2G \cong \mathbb{Z}_2^r$ has 2^r elements. An infinite basis Y would give infinitely many distinct cosets $y + 2G$ (distinct basis elements are distinct mod $2G$ by uniqueness of coefficients mod 2). ■

Theorem 7.27 (38.8)

If G is generated by $a_1, \ldots, a_n$, then $\phi : \mathbb{Z}^n \to G$, $\phi(h_1, \ldots, h_n) = \sum h_ia_i$, is an onto homomorphism.

Theorem 7.28 (38.9)

If $\{x_1, \ldots, x_r\}$ is a basis of G, $t \in \mathbb{Z}$ and $i \neq j$, then replacing x_j by $x_j + tx_i$ gives another basis.

Proof. The new set generates ($x_j = (x_j + tx_i) - tx_i$), and a relation $\sum n_kx'_k = 0$ becomes $\sum_k n_kx_k + n_jtx_i = 0$, forcing $n_k = 0$ for $k \neq i$ and then $n_i = 0$. ■

Theorem 7.29 (38.11)

If G is free abelian of rank n and $K \neq \{0\}$ a subgroup, then K is free abelian of rank $s \leq n$, and there are a basis $x_1, \ldots, x_n$ of G and positive integers $d_1 \mid d_2 \mid \cdots \mid d_s$ such that $d_1x_1, \ldots, d_sx_s$ is a basis of K.

Proof. This is the stacked-basis argument in the proof of Theorem 11.12 (Volume I, Step 1), carried out with the elementary basis changes of Theorem 7.28. ∎

Theorem 7.30 (38.12)

Every finitely generated abelian group is isomorphic to $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \cdots \times \mathbb{Z}_{m_r} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$ with $m_i \geq 2$ and $m_i \mid m_{i+1}$.

Proof. By Theorem 7.27, $G \cong \mathbb{Z}^n/K$. With the bases of Theorem 7.29, $\mathbb{Z}^n/K \cong \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_s} \times \mathbb{Z}^{n-s}$; drop the factors with $d_i = 1$. ∎

Splitting each $\mathbb{Z}_{m_i}$ into prime-power factors gives Theorem 11.12.

7.6 Free groups and presentations (§§39–40)

Motivation. How can one describe an infinite or complicated group finitely? Start with the free group, in which the generators satisfy no relations at all, and impose relations; every group arises this way. Presentations are how groups appear in topology (fundamental groups) and are the natural language for classifying small groups.

For a set A of letters, **words** are finite strings of symbols a, a^{-1} ($a \in A$); **elementary contractions** delete adjacent pairs aa^{-1} or $a^{-1}a$; every word reduces to a unique **reduced word** (uniqueness is checked by van der Waerden's permutation trick, as for free products). The reduced words form a group $F[A]$ under juxtaposition followed by reduction, the **free group** on A. A group G is **free on** $A \subset G$ if $G \cong F[A]$ by an isomorphism fixing A; the cardinality of A is its **rank**.

Theorem 7.31 (39.7)

If G is free on A and on B, then $|A| = |B|$.

Proof. The abelianization G/C (Theorem 15.20) is free abelian with basis the images of A (a homomorphism $G \to \mathbb{Z}^{(A)}$ sending a to the basis vector e_a exists by Theorem 7.34, and kills C), and likewise with basis the images of B. If A is finite, then so is B and $|A| = |B|$ by Theorem 7.26. If both are infinite, $|A| = |F[A]| = |B|$, since the set of finite words over an infinite set has the same cardinality as the set. ∎

Theorem 7.32 (39.9)

Two free groups are isomorphic iff they have the same rank.

Proof. A bijection $A \to A'$ extends to an isomorphism $F[A] \to F[A']$ (rename letters); conversely an isomorphism carries a free generating set to one, and use Theorem 7.31. ∎

Theorem 7.33 (39.10, Nielsen–Schreier)

A nontrivial proper subgroup of a free group is free.

Proof. Realize $F[A]$ as the fundamental group of a wedge of circles X, one for each $a \in A$. For $H \leq F[A]$ there is a covering $E \to X$ with fundamental group isomorphic to H; E is a graph, and the fundamental group of a connected graph is free (generated by the edges outside a maximal tree). Full proofs are in the Topology II notes, Chapter 14. (An algebraic proof uses a Schreier transversal T of H: H is free on the nontrivial elements $ta\,\overline{ta}^{-1}$, $t \in T$, $a \in A$.) ∎

Theorem 7.34 (39.12)

Let G be generated by $A = \{a_i\}$ and let $a_i' \in G'$ be arbitrary. There is at most one homomorphism $\phi : G \to G'$ with $\phi(a_i) = a_i'$; if G is free on A, there is exactly one.

Proof. Every element of G is a product of powers of the a_i (Theorem 7.6), and ϕ is determined on such products. If $G = F[A]$, define $\phi(a_{i_1}^{n_1} \cdots a_{i_k}^{n_k}) = a_{i_1}'^{n_1} \cdots a_{i_k}'^{n_k}$ on reduced words; it is well defined, and since the product of words is computed by juxtaposition and cancellation, which corresponds to valid equalities in G', it is a homomorphism. ∎

Theorem 7.35 (39.13)

Every group is a homomorphic image of a free group.

Proof. Take $A = G$ (or any generating set) and the homomorphism $F[A] \to G$ of Theorem 7.34 extending the identity of A; it is onto. ∎

A **presentation** $(x_j : r_i)$ of G consists of generators x_j and relators r_i (words) such that $G \cong F[\{x_j\}]/N$, N the least normal subgroup containing the r_i. Examples: $\mathbb{Z}_n = (x : x^n)$; $\mathbb{Z} \times \mathbb{Z} = (x, y : xyx^{-1}y^{-1})$; $D_n = (x, y : x^n, y^2, xyxy)$; $Q_8 = (x, y : x^4, x^2y^{-2}, yxy^{-1}x)$. Using Sylow theory and presentations one classifies small groups: for instance there are exactly 5 groups of order 8 ($\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$, $\mathbb{Z}_2^3$, D_4, Q_8) and 2 of order 10 ($\mathbb{Z}_{10}$, D_5).